Does Electricity Kill the Weed Seed? Species and Stage-Specific Effects of Electrical Weed Management on Seed Germination, Viability, and Vigor of Palmer Amaranth and Barnyardgrass

Sougata Roy[1]*, Ryan Hamberg[1], Muthukumar Bagavathiannan[1]

[1]Department of Soil and Crop Sciences, Texas A&M University, College Station, TX, 77843 · sroy_agronomy@tamu.edu

Introduction

- Herbicide resistance is driving demand for non-chemical weed control **(Merfield 2024)**
- Electrical weed management (EWM) drives high-voltage current through the plant; its resistance converts this to heat, rupturing cells **(Fig 1)**
- Heating follows Joule's law, $Q = I^2Rt$ (current × resistance × contact time)
- Control of standing weeds is well established; the effect on seed viability and the seedbank is not **(Schreier *et al.* 2022)**
- We tested species and stage-specific effects of EWM on seed germination, viability and vigor

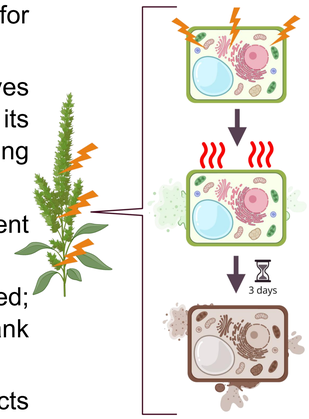


Fig 1: Mechanism of electrical energy damages in plant tissue.

Hypothesis and Objective

Hypothesis: EWM at reproductive stage reduces seed germination, viability and vigor, and the effect varies by species and seed stage.

Objective: Test how species and seed stage shape the effect of EWM on germination, viability and vigor, and how it progresses after treatment.

Materials and Methods

Experimental Design

- Field study, Summer 2026, Texas A&M AgriLife Research Farm, College Station, TX
- Factorial completely randomized design with four replications
- Treatments applied with a tractor-mounted Zasso EH60 Electroherb™ system **(Fig: 2a)** at 255 V, 18.25 A (4.65 kW, ≈ 14 Ω), 1 mph
- Seed harvested at each stage; germination %, viability, and vigor index assessed at 0 and 15 DAT **(Fig: 2b)**
- Factorial ANOVA; Tukey-adjusted pairwise comparison (α = 0.05)



Fig 2a. Tractor-mounted Zasso EH60 Electroherb™ system used to apply treatments.

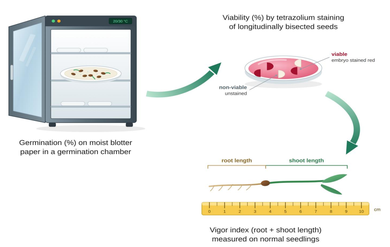


Fig 2b. Seed viability and vigor testing workflow.

Treatment Structure

2 SPECIES	3 SEED STAGES	2 REGIMES	2 TIME-POINTS
Palmer amaranth Barnyardgrass	Early · Mid · Late	EWM Untreated control	0 DAT · 15 DAT

Results and Discussion

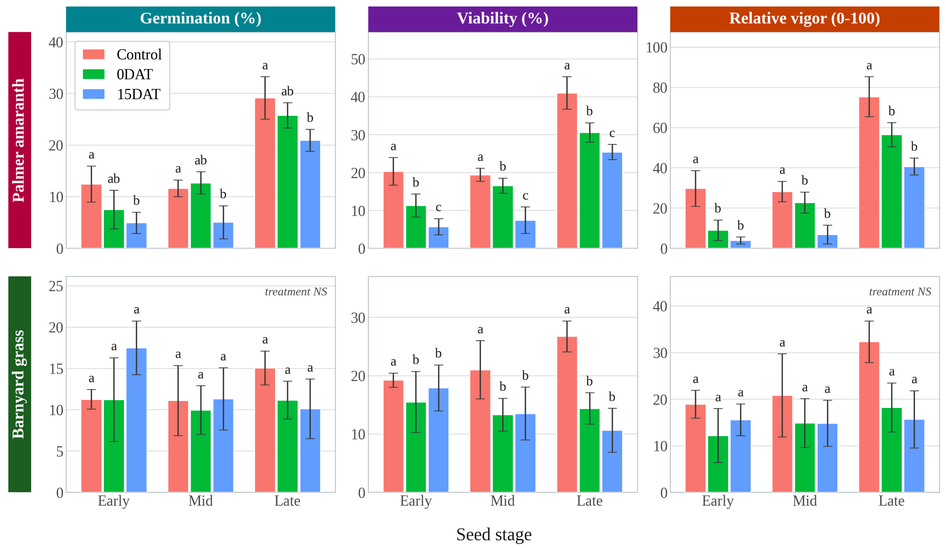


Fig 3. Germination, viability, and seedling vigor index of Palmer amaranth and Barnyardgrass by seed developmental stage and treatment. (Bars are means of four replications; bars are ± 1 SE.)

- Palmer amaranth responded strongly to EWM across all three traits; barnyardgrass showed a clear response only in viability and vigor **(Fig 3) (Schreier *et al.* 2022)**
- Early-stage Palmer amaranth germination fell from **12.4% to 4.9%,** and viability from **20.3% to 5.7%,** at 15 DAT
- Barnyardgrass germination was **not significantly** affected at any stage **(Table 1)**
- In every trait the 15 DAT bars sit below 0 DAT; damage progresses under field exposure rather than acting only at treatment
- Untreated seed gained germination, viability, and vigor with maturity; EWM suppressed that gain, so late-stage seed keeps the largest advantage **(Tangney *et al.* 2019)**

Table 1. Significance of EWM effects by species and seed trait.

	Germination	Viability	Vigor index
Palmer amaranth	0.013	< 0.001	< 0.001
Barnyardgrass	NS	0.02	< 0.001

Cell shading: dark = P < 0.001, mid = P < 0.05, grey = not significant. Factorial ANOVA, Tukey-adjusted (α = 0.05).

Applied EWM dose

- 255 V · 18.25 A
- Power = 4.65 kW
- Resistance = 13.97 Ω (soil + plant)
- Travel speed = 1 mph

Vigor index

- VI = germination % × (root + shoot) in cm
- Relative vigor = (VI ÷ 597.4) × 100

597.4 = highest VI observed



Fig 4. Seed appearance after treatment: EWM-treated versus untreated control.

Table 2. Viability reduction at 15 DAT by seed stage.

Viability	Early	Mid	Late
Palmer amaranth	72%	62%	38%
Barnyardgrass	7%	36%	60%

Conclusions

- EWM significantly reduced seed viability in Palmer amaranth, confirming that the treatment reaches the seed and not only the vegetative plant **(supports hypothesis partially)**
- The response is species-specific: Palmer amaranth germination dropped significantly while barnyardgrass germination did not, though barnyardgrass viability and vigor both fell. **(Slaven *et al.* 2023).**
- Effects were greatest on early-stage seed and increased between 0 and 15 DAT, timing of application is the operative lever.

Treating weeds early, while seeds are still forming, means fewer live seeds reach the soil, a non-chemical way to fight herbicide-resistant weeds.

Future Research

- Extend to additional species, including sorghum-system weeds, and to seed already shed onto the soil surface **(Schwartz-Lazaro *et al.* 2017)**
- Quantify how seed moisture content and electrode contact time govern seed kill, and track seedbank persistence beyond 15 DAT

References

Merfield, C.N. (2024). A comprehensive guide to electrical weed management. BHU Future Farming Centre Bulletin 2024(1), 64 pp.

Michigan State University Extension (2023). Basics of electrical weed control. East Lansing, MI.

Schreier, H., Bish, M. & Bradley, K.W. (2022). The impact of electrocution treatments on weed control and weed seed viability in soybean. Weed Technology 36(4), 481–489.

Slaven, M.J., Koch, M. & Borger, C.P.D. (2023). Exploring the potential of electric weed control: a review. Weed Science 71(5), 403–421.

Schwartz-Lazaro, L.M., Green, J.K. & Norsworthy, J.K. (2017). Seed retention of Palmer amaranth (Amaranthus palmeri) and barnyardgrass (Echinochloa crus-galli) in soybean. Weed Technology 31(4), 617–622.

Tangney, R., Merritt, D.J., Fontaine, J.B. & Miller, B.P. (2019). Seed moisture content as a primary trait regulating the lethal temperature thresholds of seeds. Journal of Ecology 107(3), 1093–1105.

Acknowledgements

Texas A&M AgriLife Research and the Department of Soil and Crop Sciences, College Station.

USDA National Institute of Food and Agriculture U.S. DEPARTMENT OF AGRICULTURE

zasso™